Name.....

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P525/1
CHEMISTRY
Paper 1
August 2026
2¾ hours

MASAKA DIOCESAN EXAMINATIONS BOARD

Uganda Advanced Certificate of Education
JOINT MOCK EXAMINATIONS 2026
CHEMISTRY
PAPER 1
(Theory)
2 hours 45 minutes

INSTRUCTIONS:

*This examination paper consists **two** sections **A** and **B**.* It has six examination items.

*Section A has **two compulsory** items.*

*Section **B** has **two** parts; **I** and **II**. Respond to **only one** item from each part.*

*Respond to **four** items in **all**.*

*Any additional item(s) responded to will **not** be scored.*

All *responses must be written on answer sheets provided.*

Begin each item on a fresh page

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SECTION A:

Attempt all items in this section

ITEM ONE;

Scenario

Background

Ammonia is useful raw material in manufacture of fertilisers but its rate of production requires optimization and careful study of factors affecting the production process.

A factory manufactures ammonia by reacting nitrogen with hydrogen according to the equation:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92kJmol^{-1}$$

The factory recently installed two reactors, **Reactor 1** and **Reactor 2**. Unfortunately, the labels indicating which reactor had an iron catalyst added during the reaction were lost during maintenance.

A process engineer collected test experimental data from the two reactors and other data from and text book to use and find out the components of each reactor and how their chosen operating conditions; addition of iron, pressure and temperature affect the rate of ammonia production.

One worker is also wondering which concentration of the two reactants should be doubled as the other is kept constant to obtain the maximum rate of ammonia production but he can't know this without the rate equation of the reaction.

Table 1

Reaction	Activation energy of reverse reaction,Eb ($kJmol^{-1}$)
Reaction 1	+730
Reaction 2	+272

Table 2

Experiment	$[N_2]$ ($moldm^{-3}$)	$[H_2]$ ($moldm^{-3}$)	Initial rate ($moldm^{-3}s^{-1}$)
1	0.20	0.20	4.0×10^{-3}
2	0.40	0.20	8.0×10^{-3}
3	0.20	0.40	1.6×10^{-2}
4	0.40	0.40	3.2×10^{-2}

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The company has invited outstanding chemistry students through its industrial partnership programme to join the process improvement team. As a member of the team, you have been selected to analyse the available data and recommend improvements to the factory.

Tasks

(a) Analyse the data to deduce which reaction was catalysed by iron and also to explain which reactant concentration should be doubled.

(b) Explaining how their chosen operating conditions affect the reaction rate and the applicability of law used in (a) in real life while recommending which reactor should be used for efficient production.

ITEM TWO;

Scenario

In chemical processes, understanding acid-base reactions and resulting pH levels is vital for safety and environmental protection.

At a science outreach center in Kampala, Katrinah accidentally knocked over a beaker, spilling **65cm³ of 0.35M nitric acid** across the workstation table. In an attempt to neutralize the spill quickly, she poured **25cm³ of 0.91M ammonia solution** onto it, following the reaction:

$$NH_3(aq) + HNO_3(aq) \rightarrow NH_4NO_3(aq)$$

She assumed the resulting solution was neutral ($pH \approx 7$), meeting the center's disposal requirement of **pH 6.5–8.5**.

However, her colleague Joram argued that the mixture remained unsafe and acidic and claimed that **26.39cm³** of the same ammonia was actually needed to reach a safe pH. He further asserted that managing chemical spills requires a deep understanding of how environmental **temperature** and **concentration** alter the pH of acid solutions, as well as an understanding of how specialized buffer solutions find critical **clinical** and **industrial use**. Their supervisor confirmed that Joram was correct but asked you, given your knowledge in chemistry, to provide the technical validation for the safety report to explain the science to the younger counterparts.

(K_b *for ammonia* $= 1.8 \times 10^{-5} moldm^{-3}$ and $K_w = 10^{-14} mol^2 dm^{-6}$ at 25°C)

Tasks

(a) Verify why Katrinah's initial mixture fails the safety requirement, while Joram's suggestion successfully passes the safety standards.

(b) Assess the scientific validity of Joram's assertion.

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SECTION B

*Attempt only **two** items from this section, **only one** from each of the parts **I** and **II**.*

PART I

ITEM 3

Scenario

A pharmaceutical manufacturing company uses **Group 2 elements** in the production of medicines and other industrial products.

A group of Senior Five (S5) chemistry students, through their science club, visited the factory to learn how these elements are selected for different applications.

During the tour, the students observed that some products were manufactured using magnesium, others using calcium, while a few specialized components were made from other Group 2 elements. The industrial chemist explained that the factory considers several physical and chemical properties of the elements before selecting them for specific uses. To support the discussion, the chemist provided the following data.

Element	Beryllium	Magnesium	Calcium	Strontium	Barium
Atomic radius (nm)	0.112	0.160	0.197	0.215	0.222
Melting point (°C)	1287	650	842	777	727

The students also learned that the manufacturing process involves reactions of the elements with water and dilute sulphuric acid, although no explanation of these reactions was provided. Furthermore, they noticed that one stage of production required a metal that could melt easily and flow into moulds to produce disposable medical components, while another application required a metal that could withstand moderately high temperatures without melting.

The learners returned to school wondering which elements would suit the different industrial applications, how the reactions with water and the acid take place, any other application, possible dangers associated with using any of the selected elements in industry and the measures that can be taken to minimise those risks.

Their chemistry teacher has referred these questions to you for explanation.

Task

Help the learners address their concerns.

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ITEM 4

Scenario

Halogens find extensive everyday applications. Their uses range from creating electrical insulation materials to exploiting their toxicity for disinfection purposes.

A company that manufactures electrical equipment and high-performance chemical products is planning to expand its production line to include a water treatment system. The company intends to use one Group 17 element in the manufacture of a high-performance polymer used for insulating electrical cables and another for disinfecting water supplied to nearby communities.

During a planning meeting, one engineer suggested using fluorine for producing the insulating material and another engineer recommended chlorine for water treatment but did not give reasons for their recommendations. However, the environmental and safety officer cautioned that both elements are highly reactive and hazardous if improperly handled.

To support the discussion, the laboratory provided the following information about the halogens.

Element	Fluorine	Chlorine	Bromine	Iodine
Boiling point (K)	85	239	332	457
Electronegativity	3.98	3.16	2.96	2.66

The research team also observed that the halogens differ in their reactions with magnesium metal, hydrogen gas, and concentrated potassium hydroxide solution, producing products with different industrial importance. The company management has requested an explanation for these differences before approving the materials for production.

In addition, the company is assessing strategies on how the hazardous nature of the suggested elements can be minimized while maintaining their industrial benefits.

You are a chemistry student on an educational visit at the company.

Task;

Write a comprehensive report explaining how the properties of the suggested elements dictate their specified uses, how their nature govern the stated chemical reactions, and what specific dangers they pose alongside practical safety measures to mitigate these hazards in the plant.

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PART II

ITEM 5

Scenario

Butan-2-ol, but-2-yne, and methyl ethanoate are volatile organic compounds processed at a solvent making plant in Banda, near a wetland ecosystem.

During a severe thunderstorm, flood cleanup at the plant near accidentally erased identification marks on two drums, both containing compounds with a molecular mass of **74 g/mol** ; **butan-2-ol** and **methyl ethanoate**. To eliminate severe safety and environmental mixing hazards, the plant supervisor, Mr. Opio, canceled a facility tour and **tasked a visiting chemistry class** with analyzing liquid samples to safely identify them. They recorded the following results;

- **Liquid X:** Boiling point 99.5°C , harsh, rubbing-alcohol odour.
- **Liquid Y:** Boiling point 57.0°C , Sweet, glue-like odour.

To help the factory secure the urgent environmental safety clearance required by regulators to resume operations, your teacher tasked you with compiling a comprehensive technical report. As the group leader, your report must formally identify Liquids X and Y, detail the reaction mechanisms of the plant's automated lines; dehydration of butan-2-ol using concentrated phosphoric acid and room-temperature complete chlorination of but-2-yne, evaluate the chlorination product which they intend to use in making heavy duty engine degreasers to, assess its ecological impact on the wetland, and outline engineering pathways to upgrade surplus butan-2-ol into methyl ethanoate and but-2-yne respectively.



Tasks;

(a) Account for the nature of liquids X and Y their variation in boiling points despite their identical molecular mass and propose the synthetic roadmaps that the engineers must evaluate.

(b) Elucidate the key chemical processes the factory undertakes and assess **two** applications, **two** environmental dangers of the highlighted process to the wetland ecosystem and propose mitigation strategies.

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ITEM 6

Scenario

The manufacture of materials like polymers and fuel additives has vastly advanced modern industry, requiring an in-depth understanding of the behavior of organic molecules.

A factory needs to produce two things: a safe plastic for making food containers and a fuel additive that prevents premature, uncontrolled combustion inside an internal combustion engine.

At a meeting, engineers suggested making polypropene for the containers and ethylbenzene for the fuel additive. They plan to use ethanoic acid and ethanol as starting chemicals to make them respectively. However, some staff worried about the dangers of leaving unused starting chemicals in the factory. To solve this and earn extra money, a manager suggested turning the leftover acid into a useful ester and converting the leftover alcohol into diethyl ether to sell.

Since the team lacks chemistry expertise on these ideas, the company has invited top science students for a technical review session through its corporate social responsibility program. You have been selected to represent your school.

Tasks

Help the company to;

(a) Account for the suitability of the suggested products for their role and how the mitigation reactions occur.

(b) While assessing the health implications of the two processes, explain how the selected products can be obtained from the suggested raw materials.

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PERIODIC TABLE

1	2											3	4	5	6	7	8
1 H 1.0																1 H 1.0	2 He 4.0
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3											13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.4	18 Ar 40.0
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc 98.9	44 Ru 101	45 Rh 103	46 Pd 103	47 Ag 108	48 Cd 112	49 In 115	50 Sn 119	51 Sb 122	52 Te 128	53 I 127	54 Xe 131
55 Cs 133	56 Ba 137	57 La 139	72 Hf 178	73 Ta 181	74 W 184	75 Re 186	76 Os 190	77 Ir 192	78 Pt 195	79 Au 197	80 Hg 201	81 Ti 204	82 Pb 207	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)															

	57 La 139	58 Ce 140	59 Fr 141	60 Nd 144	61 Pm (145)	62 Sm 150	63 Eu 152	64 Gd 157	65 Tb 159	66 Dy 162	67 Ho 165	68 Er 167	69 Tm 169	70 Yb 173	71 Lu 175
	89 Ac (227)	90 Th 232	91 Pa 231	92 U 238	93 Np 237	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf 251	99 Ea (254)	100 Fm (257)	101 Mv (256)	102 No (254)	103 Lw 260

1

1. H – indicates Atomic number

2. H – indicates relative Atomic mass

1.0

END

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